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M-RAV: Multimodal retrieve-augment-verify framework for boosting zero-shot fact verification system with large language models

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ABSTRACT

The rapid spread of multimodal content, combining text and images, has fueled the rise of misleading information online, making multimodal fact-checking increasingly vital. However, large language models (LLMs) often struggle with cross-modality reasoning to verify claims directly. To address this, we introduce M-RAV, a portable multimodal fact-checking framework that enhances LLMs' reasoning capabilities. M-RAV operates in four key steps: it **M**ultimodally **R**etrieves relevant evidence, **A**ugments alignment between text and images by explaining their consistency, assigning the relevance score for each evidence explanation, and **V**erifies claims using this augmented evidence. By clearly explaining the relationship between text and image evidence in a way that guides LLMs, M-RAV significantly improves fact-verification accuracy. Evaluated on three benchmark multimodal fact-checking datasets, M-RAV demonstrates efficiency even in zero-shot settings. In gold-evidence scenarios, M-RAV outperforms state-of-the-art methods by approximately 3% on Mocheg, 5% on Fin-Fact, and delivers competitive results with fine-tuned models on FACTIFY. In system-retrieved evidence scenarios, M-RAV boosts performance by nearly 6% compared to the best existing method, demonstrating its potential for practical, real-world fact-checking systems.

1. Introduction

Fact-checking verifies the accuracy of claims, particularly those shared on social media, which may contain truthful or misleading information (Guo, Schlichtkrull, & Vlachos, 2022). With the rapid spread of information on social networks, fact-checking is critical but time-consuming (Guo et al., 2022; Vlachos & Riedel, 2014). The challenge intensifies as users access information quickly, making it difficult for administrators to detect and verify misinformation. Automated fact-checking (AFC) addresses this by modeling the process as a Natural Language Processing (NLP) task, involving evidence retrieval and claim verification (Guo et al., 2022). AFC reduces the impact of misinformation online, but claims and evidence often appear in multiple modalities — text, images, or videos — posing additional challenges (Akhtar, et al., 2023).

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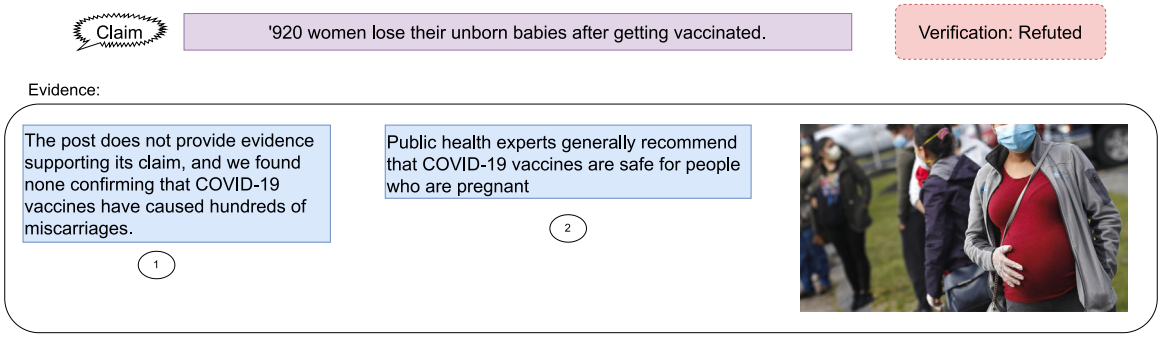


Fig. 1. Example of a claim with text and image evidence. The example is extracted from the Mocheg dataset.

Multimodal fact-checking is increasingly vital due to the prevalence of misleading content combining text and images (Akhtar, et al., 2023). However, it faces two main obstacles. First, training fact-checking models requires large, human-annotated datasets, which are costly, time-consuming, and prone to bias (Pan, Wu, et al., 2023). Second, models must reason across modalities, integrating text and visual information to identify features that support or refute a claim (Yao, Shah, Sun, Cho, & Huang, 2023). For example, consider the claim: *"920 women lose their unborn babies after getting vaccinated"*. As shown in Fig. 1, evidence includes text stating, *"There is no confirmation about miscarriages",* and *"Public health experts suggest COVID-19 vaccines are safe for pregnant women",* alongside an image of a pregnant woman wearing a mask during the COVID-19 pandemic.[2] The evidence suggests the claim is false, but the image alone cannot confirm vaccine effects, and manipulated visuals can introduce bias (Akhtar, et al., 2023). Thus, effective multimodal fact-checking requires combining text and image evidence clearly and reliably.

To address these challenges, large language models (LLMs) like GPT-4 offer a promising solution. LLMs store vast knowledge and excel at reasoning, enabling claim verification with minimal or no training data (Geng, Kementchedjhieva, Nakov, & Gurevych, 2024; Nakov, et al., 2021). Their reasoning capabilities support logical deductions based on claims and evidence, improving verification accuracy (Chen, Liu, & Zhou, 2025). Prompting techniques, such as zero-shot or few-shot prompting, leverage LLMs' knowledge in data-scarce scenarios (Vykopal, Pikuliak, Ostermann, & Šimko, 2024). However, ambiguous prompts can reduce performance, necessitating clear, precise instructions. Additionally, establishing explicit connections between claims and multimodal evidence is crucial, as discrete evidence (e.g., text and images in Fig. 1) complicates verification. To sum up, we demonstrate two main challenges for multimodal fact verification as follows.

Challenge 1: Cross-modality aligning between text and image

The heterogeneity between text and image reduces the accuracy of the model because the information from each modality, such as text or image, can overlap and be redundant, which leads to inconsistency and complicates features (Fig. 1 illustrates an example of the confusing between text and image evidence in verifying a claim). Addressing the heterogeneity helps improve the performance of the model through multimodal integration and reasoning (Li, et al., 2025). To tackle this problem for the multimodal Fact-checking task, we form the first research question as **RQ1**: Does aligning text and image evidence enhance LLM reasoning for claim verification?

Challenge 2: Fact-verification with system-retrieved evidence

In the end-to-end Fact-checking system, the retrieval stage provides curated evidence for verifying the claim (Akhtar, et al., 2023). The evidence retrieval plays a vital role in providing sufficient knowledge to verify the truthfulness of the claim, since most fact-checking methods use external retrieved evidence to verify the claim (Zheng et al., 2024). In addition, most of the current multimodal Fact-checking datasets like Fin-Fact (Rangapur, Wang, Jian, & Shu, 2025) and FACTFY (Mishra, et al., 2022) consult the evaluation results on gold evidence only. Hence, we address this challenge to investigate the performance of the fact-checking model with the system-retrieved evidence scenario. We form this problem as the second research question: **RQ2** How effective is the fact-checking system with automatically retrieved evidence in practical, unsupervised fact-checking?

We address these challenges by proposing M-RAV, a multimodal fact-checking framework that harnesses LLMs to verify claims using text and image evidence. M-RAV operates in four stages: (1) Retrieve relevant evidence using the claim as a query, (2) Augment evidence by aligning text and image content for consistency, (3) Scoring the relevance between the claim and aligned evidence, and (4) Verify the claim's truthfulness by assessing evidence relevance. By employing zero-shot or few-shot prompting, M-RAV minimizes reliance on annotated data (Vykopal et al., 2024). It supports both manually curated (gold) and automatically retrieved (system) evidence, making it adaptable to real-world scenarios. Our contributions are summarized as:

- We introduce M-RAV, a novel end-to-end framework for multimodal fact-checking across text and image modalities.
- We introduce an alignment mechanism that enhances cross-modal consistency between textual claims and visual evidence by leveraging the explanation of the consistency between textual and visual evidence data.

[2] The image originates from a 2021 Washington Post article about pregnant women during the pandemic: https://www.washingtonpost.com/health/2021/08/11/pregnant-covid-vaccine/.

- We develop a relevance scoring strategy to choose the most appropriate evidence for the claim that helps improve the performance of the system under noisy retrieval conditions.
- We demonstrate that M-RAV is efficient in the practical application of fact-checking system development through evaluating M-RAV on two scenarios: gold evidence provided by three multimodal fact-checking datasets and the system-retrieved evidence, where the evidence is automatically retrieved by the system and used for verifying the claim.

The paper is organized as follows: Section 2 reviews fact-checking with language models. Section 3 details the M-RAV methodology. Section 4 analyzes three benchmark datasets. Section 5 presents experimental results and discusses challenges. Section 7 summarizes the work.

2. Related works

Since Fact-checking is a task that verifies and detects the truthfulness of information, there are several associated tasks to fact-checking, like fake news detection, disinformation, and misinformation detection. However, the scope of these tasks is different from fact-checking, as the fake news detection task focuses on assessing the aspect of information about the satire or hate speech, which is not relevant to the veracity (Guo et al., 2022; Thorne & Vlachos, 2018), and the misinformation detection focuses on detecting the aspect of information that is not accurate or complete (Thorne & Vlachos, 2018). Similarly, the disinformation detection task is designed to detect whether the information contains malicious motives to cause misleading to the audience rather than verify its veracity (Thorne & Vlachos, 2018). Therefore, the previous works in Akhtar, et al. (2023), Guo et al. (2022), Thorne and Vlachos (2018) use the term fact-checking or fact-verification to describe the process of detecting the veracity of the claim. Recent studies on misinformation detection further show the growing use of transformer-based neural architectures for assessing unreliable online content. For example, Khan and Chen (Khan & Hongsong, 2025) propose hybrid transformer deep neural architectures for misinformation detection on social media. Although such work is related to our motivation, it differs from fact-checking because it focuses on detecting misleading content patterns rather than verifying a claim against retrieved multimodal evidence. Fact-checking systems verify a claim by comparing it with evidence from reliable sources. One common approach measures the semantic similarity between the claim and vetted documents, as in FacTeR-Check (Martín, Huertas-Tato, and Huertas-García, Álvaro and Villar-Rodríguez & and Camacho, 2022) and VClaim (Mansour, Elsayed, & Al-Ali, 2023). Another line of work uses natural-logic inference to test whether the evidence supports or contradicts the claim, exemplified by ProoFVer (Krishna, Riedel, & Vlachos, 2022) and Zero-NatVer (Strong, Aly, & Vlachos, 2024). As information becomes more complex, many claims require assembling multiple pieces of evidence. To handle this, recent methods model relationships among claims, evidence sentences, and sources using graphs — often via Graph Neural Networks or Knowledge Graphs — thereby capturing structured connections that improve verification, as shown by HHGN (Chen, Cai, Hu, Chen, & Chen, 2021), HybridFC (Qudus, Röder, Saleem, & Ngonga Ngomo, 2022), and Traceable LLM (Adam & Kliegr, 2025). Additionally, the work by Peng, Jian, Kan, Qiao, and Li (2024) demonstrates that fake news is semantically diverse across modalities and proposes contextual semantic representations to capture cross-modal relationships, while Li, et al. (2025) addresses modal heterogeneity through uncertainty-aware disentangled representation learning. Other works like DSMM (Ma, Hu, Ge, & Zhang, 2024) and NSEP (Fang, et al., 2024) present advanced network paradigms, such as position-aware mechanisms, graph-based, attention-based networks, and multi-task learning, that efficiently exploit contextual information surrounding the claim as evidence to determine its veracity. However, these methods rely heavily on supervised training with large annotated datasets, which is a limitation in real-world applications where annotated data is unavailable and the dynamic change in social data.

Large Language Models (LLMs) have become powerful tools for fact-checking due to their ability to process and evaluate large amounts of text with high accuracy. Recent work on fake-news and harmful content governance in the LLM era also emphasizes the need for reliable, controllable, and trustworthy language-based systems for online information assessment (Ahmad & Hongsong, 2026; Anjum, Latif, & Chen, 2026). Zero-shot fact-checking is particularly promising, as it allows LLMs to verify claims without needing time-consuming and costly annotated training data. In this approach, LLMs analyze a claim alongside relevant evidence to determine its truthfulness. To enhance LLM performance, researchers guide the models with structured reasoning steps, such as Chain-of-Thought (Wei et al., 2022), or use semantic representations. Additionally, ProgramFC (Pan, Wu,et al., 2023) decomposes claims into sub-tasks, verifying each against a knowledge source using LLMs. Min, et al. (2023) propose FactScore, which breaks complex claims into smaller atomic facts and compares them to reliable sources. Recently, several fact-checking methodologies have leveraged the reasoning ability of LLMs in retrieving and verifying information from users' queries, such as Yuan and Vlachos (2024) represent claims and evidence as semantic triplets using Knowledge Graphs to aid verification, Chen et al. (2025) demonstrates the promise of LLMs for preemptive rumor detection using accessible social context, ZeFAV (Luu, Nguyen, Vo, & Nguyen, 2024) transforms claims and evidence into relational triplets and combines them with restructured evidence documents using InfoRE (Cheng, Tan, Xue, & Lu, 2024), a method that organizes documents compactly to improve reasoning, DEFAME (Braun, Rothermel, Rohrbach, & Rohrbach, 2025) employs a dynamical selection in search tools to retrieve and evaluate textual and visual evidence for verification by Multimodal LLMs, and FM-KBQA (Gao, Cao, Bu, Zhu, Guan, & Yu, 2025) instructs LLMs to generate logical forms for reasoning to exploit the answer. While these methods outperform supervised models, they focus solely on text-based evidence and are less suited for multimodal (text and image) content.

Multimodal fact-checking addresses claims involving both text and images, requiring methods to integrate and align these data types. Recent approaches have achieved strong results. For example, Lee, et al. (2024) use external LLMs (e.g., GPT-4, GPT-3.5) to explain the context between images and text, enhancing verification. Geng et al. (2024) employ in-context learning with samples

generated by GPT-4V to guide a system (LLaVA) in verifying new claims. Cekinel, Karagoz, and Çöltekin (2025) use a Vision Language Model (VLM) to combine image and text information into feature vectors, which are then processed by a feed-forward network to predict a claim's truthfulness. The key to effective multimodal fact-checking lies in aligning image and text evidence to uncover consistent information. However, methods like Cekinel et al. (2025) require significant computational resources, making them less efficient for zero-shot settings. Additionally, the explanations in Geng et al. (2024), Lee, et al. (2024) do not emphasize consistency between images and text, a critical factor for accurate verification. These methods also rely on commercial LLMs (e.g., GPT-4, GPT-4V), which depend on costly API services and are less practical for independent systems (Pan, Lu, et al., 2023).

Our work addresses these limitations by proposing an efficient zero-shot fact-checking method that leverages open-source LLMs. By avoiding reliance on commercial APIs, our approach is designed for practical, standalone systems. We focus on generating consistent alignments between text and image evidence to enhance verification, improving performance in multimodal settings without excessive computational demands.

3. Method

The M-RAV framework is a zero-shot learning approach designed to verify the truthfulness of claims using large language models (LLMs) and multimodal evidence (text and images). Inspired by fact-checking pipelines (Guo et al., 2022), M-RAV aims to create a practical, portable system for real-world applications. Given a claim, the framework retrieves relevant evidence, aligns text and image evidence, evaluates their relevance, and uses LLMs to determine the claim's veracity. The process consists of four stages, as illustrated in Fig. 2. Each stage is described as follows:

Stage 1: Evidence Retrieval Given a claim c, the system retrieves relevant evidence from a database $\mathcal{DB}$, built from fact-checking platforms like Snopes[3] and PolitiFact.[4] We use CLIP (Radford, et al., 2021) to fetch the top-k images $I = \{img_1, \dots, img_n\}$ and FlagEmbedding BGE (Chen, et al., 2024) to retrieve the top-k texts $T = \{text_1, \dots, text_m\}$, where n and m depend on the chosen k. The optimal value of k is discussed in Section 5. In our implementation, $\mathcal{DB}$ consists of texts and images from the Mocheg database constructed on top of PolitiFact and Snopes (Yao et al., 2023). There are 82,206 text articles and 122,186 images in $\mathcal{DB}$. The empirical study for choosing the efficient top-k is described in Section 5.4.

Stage 2: Evidence Alignment This stage aligns text and image evidence to extract meaningful connections. For each pair of image $img_i \in I$ and text $text_j \in T$, we create a set of pairs $P = \{(img_i, text_j), \dots\}$, covering all $n \times m$ combinations. An LLM generates a short explanation for each pair, describing whether the image and text are consistent, using the following prompt template:

Evidence Alignment Prompt Template

```
Please generate a short paragraph describing the consistency of the image based on the given text,
    following this template:
   <HYPOTHESIS>: Please determine whether the image is consistent with the text or not.
   <EXPLANATION>: Explain the alignment between the image hypothesis and the text.
   <FINAL ANSWER>: Give one paragraph describing the consistency of the image and text based on the
    explanation.
   <RESPONSE>:
```

The output is a list P', where each element is a triple (h, e, c): h states the hypothesis (consistent or not), e provides the explanation, and c summarizes the consistency. This verbalized alignment, as shown in Fig. 2, enhances the LLM's ability to reason about multimodal evidence. We implement the alignment step with *Llama-3.2-90B-Vision-Instruct*, which ingests each $(img_i, text_j)$ pair and produces the structured triple (h, e, c) as specified above.

Stage 3: Relevance Scoring To assess the relevance of evidence to the claim, we compute similarity scores between the claim c and each verbalized evidence pair in P'. We use LLM2Vec (BehnamGhader, et al., 2024) to obtain sentence-level embeddings by reusing a large language model as an encoder. In our setup, we embed the claim c and the concatenated verbalization v_{ij} for each pair and compute cosine similarity. LLM2Vec has been reported to provide strong text embeddings across tasks (Cao, 2024), making it a practical choice for robust relevance scoring.

Stage 4: Claim Verification In the final stage, the LLM verifies the claim's truthfulness based on the claim c and the evidence set E (including text, images, alignment explanations, consistency conclusions, and relevance scores). The LLM outputs a label $lb \in$ {supported, refuted, not enough information}, treating the task as a multiclass classification problem. The following prompt guides the LLM:

[3] https://www.snopes.com/
[4] https://www.politifact.com/

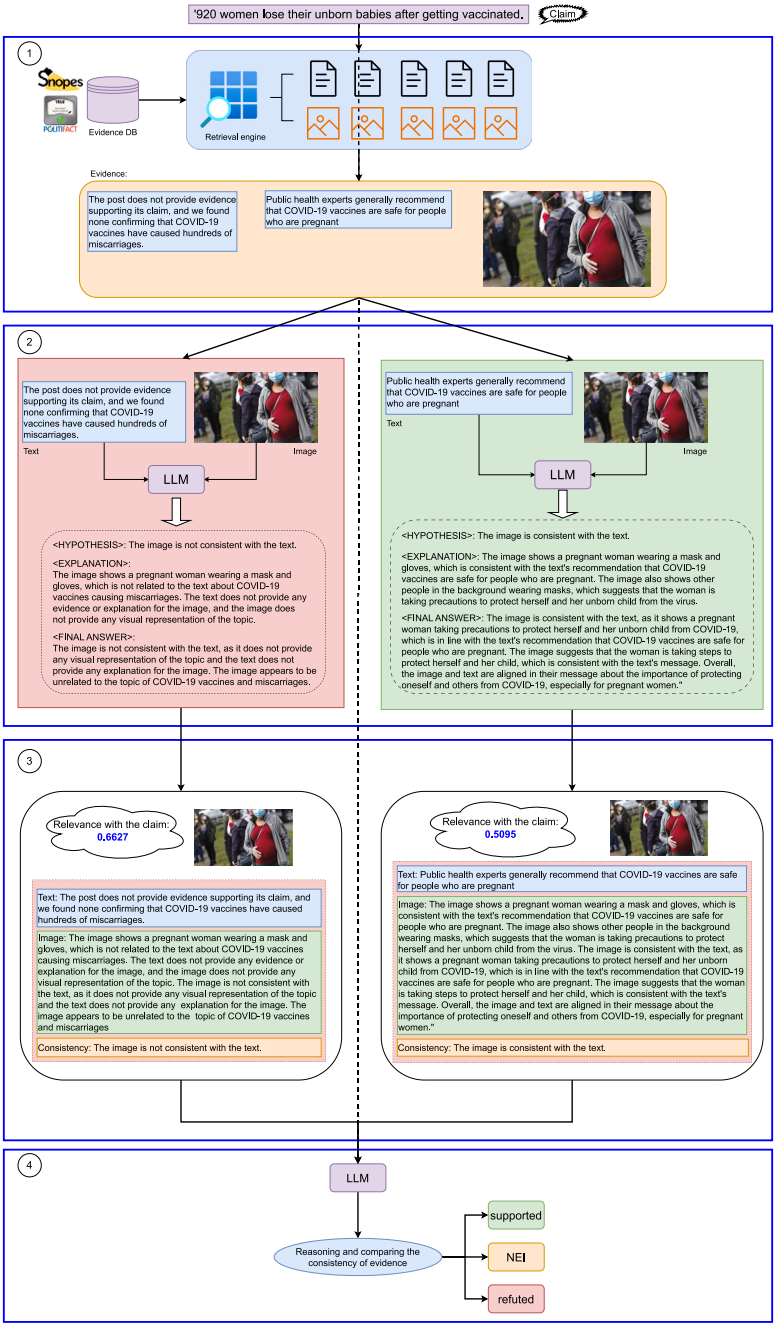


Fig. 2. Overview of the M-RAV framework.

Claim Verification Prompt Template

```
Is it true that: <the claim>?
Here is the evidence for checking:
    Text: <text evidence>
    Image: <image evidence>
    Description: <the explanation for the alignment between the text and image evidence>
    Consistency: <the conclusion about the consistency between the text and image evidence>
    Relevance score: <relevant score of claim and evidence>
    ....
To verify the truthfulness of the claim, please follow these steps:
    STEP 1: Consult the relevance between the claim and each given evidence based on the relevance
  score.
    STEP 2: Think and conclude the truthfulness of the claim based on the relevance and logic of
 the evidence.

    The truthfulness must be only one of three values: supported, refuted, or not enough
 information. Please think step-by-step carefully and response only the truthfulness of the
 claim.
<RESPONSE>:
```

From the prompt template, the "Description" indicates the explanation for the alignment between the text and image evidence, along with "Consistency" showing whether the text and image in the evidence are consistent or not. This information is extracted from Stage 2 in the M-RAV framework as shown in Fig. 2. Finally, the "Relevance score" describes the level of similarity between the claim and the evidence, which is extracted from Stage 3. A visual depiction of the operation of our M-RAV strategy on the Mocheg benchmark, with the additive information of each component, is displayed in Fig. 2. We provide a corresponding algorithmic description in Algorithm 1.

Algorithm 1 The algorithm of the M-RAV framework

1: **Input:** Claim C
2: **Retrieval** (Stage 1): Using CLIP to retrieve top-k images $\mathcal{I}$ and FlagEmbeddingBGE to retrieve top-k relevant texts $\mathcal{T}$ from $\mathcal{DB}$.
3: **Augmentation** (Stage 2):
4: **for** text t in $\mathcal{T}$ **do**
5: **for** image i in $\mathcal{I}$ **do**
6: Generate an explanation e to describe the consistency between t and i with the LLM.
7: Store the explanation e, image i, and text t. Denote this triple as raw consistent explanation p such that $p = \langle e, i, t \rangle$.
8: **end for**
9: **end for**
10: **Relevance Computation** (Stage 3): Compute the semantic similarity for all triples $\{p_j\}_{j=1}^{\|\mathcal{I}\|*\|\mathcal{T}\|}$. Denote the tuple as raw consistent text explanation ex such that $ex_j = \langle e_j, t_j \rangle$, we compute the semantic similarity to C by using LLM2Vec with the distance function as the cosine distance s_{c_j} as follows:

$$s_{c_j} = 1 - \frac{\langle ex_j, C \rangle}{\|ex_j\| \|C\|}$$

Denote this quadruple as the evidence v_j such that $v_j = \langle s_{c_j}, e_j, i_j, t_j \rangle$.
11: **Verification** (Stage 4): A function $inference_LLM$ takes $\langle C, v_j \rangle$ as an input, and generates a final label $lb \in \{\texttt{refuted}, \texttt{nei}, \texttt{supported}\}$. Mathematically, the function $inference_LLM$ representing the **Verification** module, is expressed as follows:

$$lb = inference_LLM(C, v_j)$$

4. Datasets

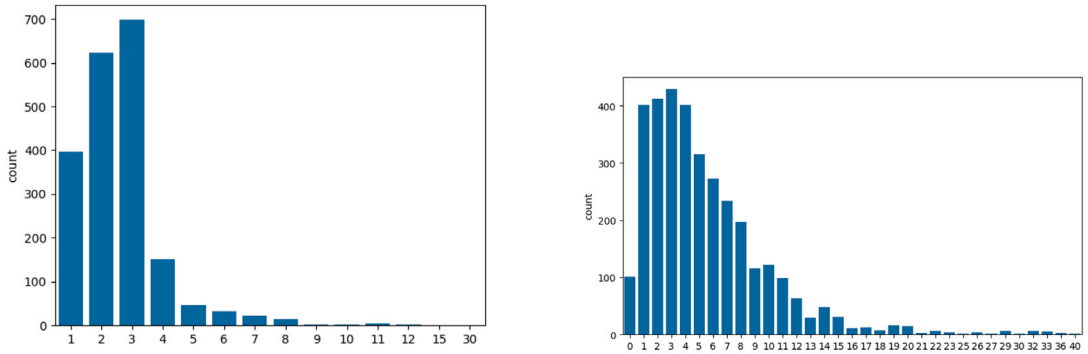
To evaluate the M-RAV framework, we use three large-scale multimodal datasets designed for fact-checking: Mocheg, FACTIFY, and Fin-Fact. These datasets, sourced from reliable platforms like PolitiFact and Snopes, are manually annotated by humans. Since M-RAV operates in a zero-shot setting, we use these datasets solely for testing, without fine-tuning the model. Each dataset has unique characteristics, making them suitable for assessing the framework's ability to handle diverse fact-checking challenges.

Mocheg: The Mocheg dataset supports end-to-end fact-checking on social media platforms like Twitter (Yao et al., 2023). Built from PolitiFact and Snopes, it includes tasks for evidence retrieval, claim verification, and truthfulness justification. We use its test set, containing 2001 claim instances, to evaluate M-RAV and compare it with state-of-the-art (SOTA) methods.

FACTIFY: FACTIFY is a dataset focused on fact-checking social news, primarily from Indian sources like Vishwas, Times of India, AFP India, and BOOM (Mishra, et al., 2022). It is designed specifically for claim verification. We use its test set, with 7500 claim instances, to assess M-RAV's performance and compare it with other approaches.

Table 1
Distribution of evidence in three datasets.

	Mocheg	FACTIFY	Fin-Fact
Max text evidence	30	1	40
Average text evidence	2.61	1	5.44
Median text evidence	2	1	4
Min text evidence	1	1	0
Max image evidence	8	1	18
Average image evidence	0.82	1	0.88
Median image evidence	1	1	1
Min image evidence	0	1	0



(a) Mocheg (b) Fin-Fact

Fig. 3. Distribution of text evidence per claim in Mocheg and Fin-Fact dataset.

Fin-Fact: Fin-Fact targets fact-checking financial information on social platforms (Rangapur et al., 2025). Sourced from PolitiFact, Snopes, and FactCheck, it supports claim verification and truthfulness justification. We use the entire dataset, comprising 3369 claim instances, to test M-RAV's performance across different large language models (LLMs).

The datasets differ in structure and challenges. Mocheg and Fin-Fact provide multiple pieces of evidence per claim, while FACTIFY offers a single piece of evidence per modality (text or image). However, FACTIFY's text evidence is significantly longer than that in Mocheg or Fin-Fact. Additionally, Mocheg often lacks sufficient image evidence, while Fin-Fact includes claims with missing image or text evidence. These unique characteristics test the robustness of M-RAV and the ability of LLMs to verify claims effectively. In the following section, we analyze each dataset's characteristics, including evidence distribution, text evidence length, claim length, and label distribution, to further understand their impact on fact-checking performance.

4.1. Distribution of evidence

According to Table 1, it can be seen that the FACTIFY dataset has a single evidence for each modality per claim. For Mocheg, there is a claim with the maximum number of text evidence equal to 30. This is similar to the Fin-Fact with a maximum of 40 pieces of text evidence for a claim. For image evidence, the maximum number of evidence per claim on Mocheg and Fin-Fact is 8 and 18, respectively. This indicates the challenge of multiple pieces of evidence in the claim on Mocheg and Fin-Fact. On average, the number of text evidence in Fin-Fact is more than in the Mocheg dataset (2.61 pieces of text evidence per claim for Mocheg and 5.44 pieces of text evidence per claim for Fin-Fact), while the number of images is nearly the same (approximately 1 image per claim on both datasets) (see Fig. 3).

Investigating the distribution of text evidence per claim, it can be seen that most of the claims in Mocheg and Fin-Fact have about 3 to 4 pieces of evidence supporting the veracity of the claim, as shown in Table 1. There are no claims with missing text evidence in the Mocheg, while the Fin-Fact has about 100 claims with missing text evidence. For the image evidence distribution, as shown in Fig. 4, most of the claims in Mocheg and Fin-Fact have about 1 to 2 image evidence. Specifically, the claim with no image in both datasets is significantly high.

4.2. Distribution of text evidence length

Table 2 shows the text evidence length distribution of Mocheg, FACTIFY, and Fin-Fact datasets. We utilize the spaCy[5] for tokenizing the text into tokens, and computing statistics based on the token level. It can be seen that FACTIFY has the longest

[5] https://spacy.io/

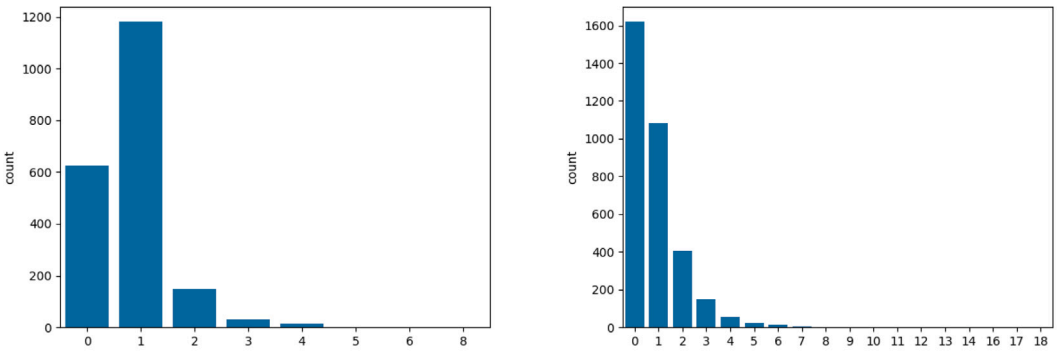


Fig. 4. Distribution of image evidence per claim in Mocheg and Fin-Fact dataset.

Table 2
Length distribution of text evidence.

	Mocheg (gold evidence)	FACTIFY	Fin-Fact	Mocheg (system evidence)
Max length	1745	55,459	2882	142,214
Average length	67.06	1107.63	63.27	481.67
Median length	33	524	51	70
Min length	5	3	0	4

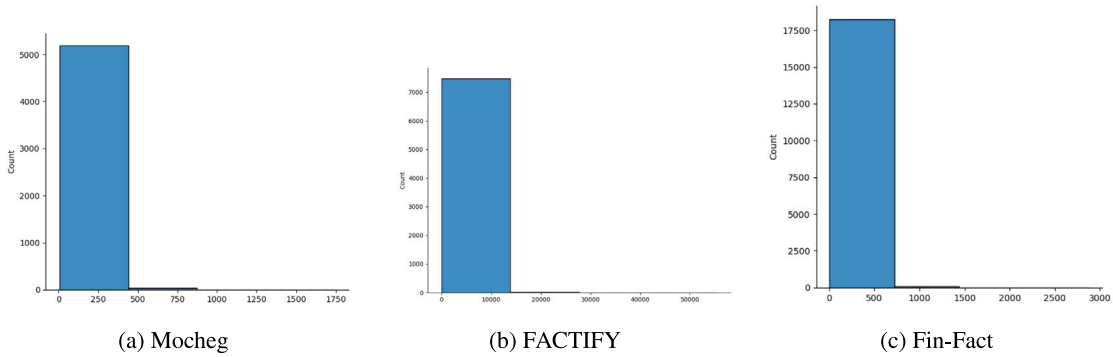


Fig. 5. Distribution of text evidence length.

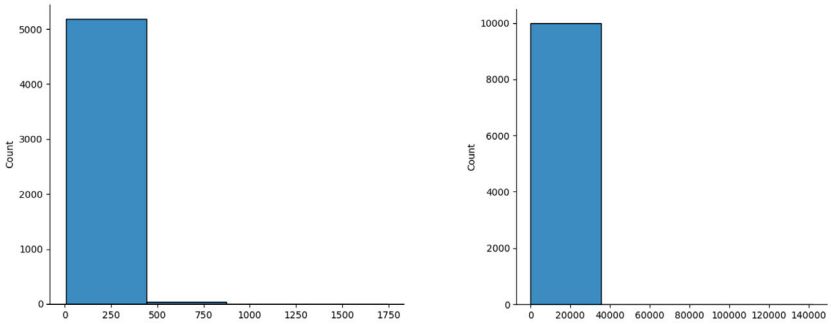
text evidence than Mocheg with gold evidence and Fin-Fact. As shown in Fig. 5, most of the text evidence in Mocheg has a length of around 250 tokens, and around 500 tokens for the Fin-Fact. Instead, most of the text evidence in FACTIFY is quite long, which is about 10,000 tokens.

Besides, as shown in Fig. 6, there is a significant difference between the gold evidence and the system evidence in the Mocheg, where most of the system evidence is much longer than the gold evidence, since the system evidence retrieved by the retrieval model is at the article level.

4.3. Distribution of claim length

Table 3 describes the maximum and minimum length of claims in three datasets. We also use spaCy[6] for tokenizing the text into tokens and computing statistics based on the token level. It can be seen that the length of the claim in Mocheg and Fin-Fact is similar, while the claim in the FACTIFY dataset is slightly longer. As shown in Fig. 7, most of the claim in Mocheg and Fin-Fact has a length of around 20 tokens, while the claim in FACTIFY contains around 50 tokens. In comparison with the text evidence, the number of tokens in the claim is much less than the evidence.

[6] https://spacy.io/

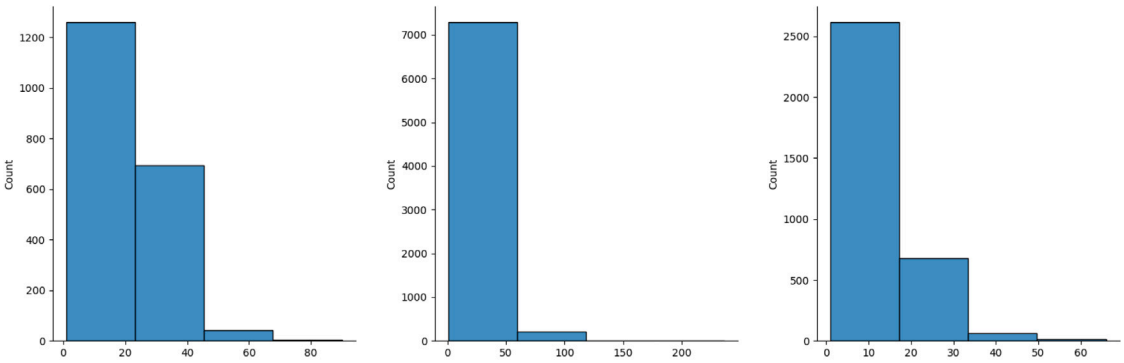


(a) Mocheg - gold evidence (b) Mocheg - system evidence

Fig. 6. Distribution of text evidence length on Mocheg.

Table 3
Length distribution of claim.

	Mocheg	FACTIFY	Fin-Fact
Max length	90	236	66
Average length	22.38	30.71	13.59
Median length	21	28	13
Mode length	16	19	12
Min length	1	1	1



(a) Mocheg (b) FACTIFY (c) Fin-Fact

Fig. 7. Distribution of claim length.

Table 4
Distribution of label in Mocheg, FACTIFY, and Fin-Fact.

	Mocheg	FACTIFY	Fin-Fact
supported	667	3000	1275
NEI	667	3000	602
refuted	667	1500	1492

4.4. Distribution of labels

Table 4 and Fig. 8 illustrate the label distribution in the three datasets. For Mocheg and FACTIFY, the statistic is conducted on the test set. Since Fin-Fact does not split the data into train, dev, or test sets, we use the whole dataset to analyze and test our methods. In general, the Mocheg has a balanced distribution on three labels, while FACTIFY takes a minority proportion at the "refuted" label, and Fin-Fact has fewer samples in the "NEI".

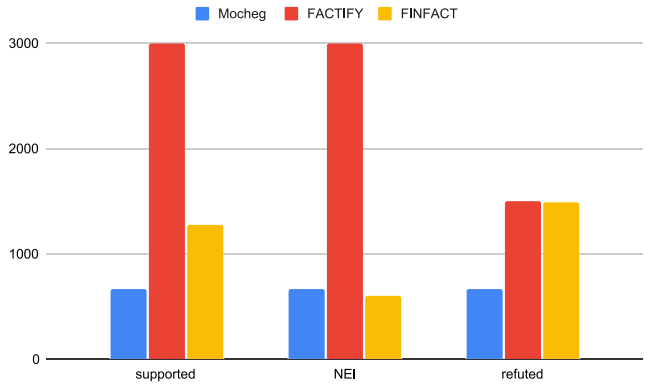


Fig. 8. Visualization of the distribution of labels in Mocheg, FACTIFY, and Fin-Fact.

Table 5
Model configurations. The *Pre-trained models* refer to the model available for download from the HuggingFace platform. In the number of parameters, M stands for Milion, and B stands for Bilion.

Task	Model name	Pre-trained models	#Parameters	Configuration
Making aligment	LlaMa 3.2 Vision	meta-llama/Llama-3.2-90B-Vision	90B	max_new_tokens=512, temperature=0.2, top_p=0.5, do_sample='True
Relevance score computation	LLM2Vec	McGill-NLP/LLM2Vec-Meta-Llama-3-8B-Instruct-mntp	3B	default configuration
Retrieval	FlagEmbedding	BAAI/bge-m3 (text evidence)	560M	default configuration
	CLIP	ViT-L/14@336px (image evidence)	306M	
Verification	LlaMa 3.2	meta-llama/Llama-3.2-3B-Instruct	3B	do_sample=False, max_new_tokens=20
	LlaMa 3.1	meta-llama/Llama-3.1-8B-Instruct	8B	
		meta-llama/Llama-3.1-70B-Instruct	70B	
	Qwen-2.5	Qwen/Qwen2.5-32B-Instruct	32B	
		Qwen/Qwen2.5-72B-Instruct	70B	
	Mixtral	mistralai/Mixtral-8 × 7B-Instruct-v0.1	56B	
	LLaVaNext	llava-hf/llava-v1.6-vicuna-7b-hf	7B	
		llava-hf/llava-v1.6-vicuna-13b-hf	13B	
		llava-hf/llava-v1.6-34b-hf	34B	

5. Empirical results

5.1. Empirical preparation

Table 5 shows the pre-trained LLMs and their configurations used in our empirical study about the M-RAV. We run the LLMs with a 4-bit quantization setting using the *BitandBytes*[7] library. All models are run using the HuggingFace library. We run the experiment on a single NVIDIA RTX A6000 with 49 GB of memory. To assess the performance of the retrieval task, we use the F1@k and MAP metrics based on the ground truth provided from the Mocheg dataset (Yao et al., 2023). The F1@K metric combines precision and recall into a single score, measuring the proportion of correctly identified relevant items relative to both the system's predicted positives and the actual ground truth positives (Huang, Chen, et al., 2024). This metric offers an efficient way to evaluate the performance of retrieval systems in practical applications. To further assess both the presence and ranking of relevant items, we incorporate Mean Average Precision (MAP) alongside F1@K, providing a comprehensive evaluation of the evidence retrieval task's performance. For the verification task, we use both micro and macro F1 scores to evaluate the performance of various models based on the gold labels in three benchmark datasets (described in Section 4).

5.2. Gold-evidence results

In this experiment, we investigate the ability of different LLMs to verify the claim. In this scenario, the gold evidence means that the claim has a set of evidence that has been created and verified by humans. We run the evaluation by two categories of LLMs: the text-only LLMs, including LLaMA (Touvron, et al., 2023), Qwen (Yang et al., 2024), and Mixtral (Albert Jiang et al., 2024), and

[7] https://pypi.org/project/bitsandbytes/

Table 6
Empirical results on gold evidence.

	Mocheg		FACTIFY		Fin-Fact	
	F1-micro	F1-macro	F1-micro	F1-macro	F1-micro	F1-macro
llama-3.2-3B + M-RAV	32.28 ↓	25.76 ↓	**39.12**↑	**33.36**↑	**29.77**↑	**26.11**↑
llama-3.2-3B	**32.43**	**27.51**	38.30	27.54	29.71	20.99
llama-3.1-8B + M-RAV	**49.92**↑	46.12 ↓	**66.86**↑	**69.54**↑	**49.24**↑	**38.23**↑
llama-3.1-8B	49.72	**49.14**	64.94	68.73	38.79	29.73
Qwen-2.5-32B + M-RAV	**57.37**↑	**56.77**↑	**66.86**↑	**70.13**↑	**49.33**↑	**48.60**↑
Qwen-2.5-32B	53.47	49.82	65.22	67.54	45.71	43.99
llama-3.1-70B + M-RAV	**55.52** ↑	**54.19** ↑	**72.33**↑	**74.87**↑	**63.52**↑	**57.23**↑
llama-3.1-70B	55.27	52.49	71.22	74.44	61.14	59.61
Mixtral-8x-7b + M-RAV	56.77 ↓	56.03 ↓	**60.06**↑	**59.79**↑	36.12 ↓	35.40 ↓
Mixtral-8x-7b	**57.97**	**57.58**	54.86	51.02	**44.43**	**43.85**
LLaVaNext-7B + M-RAV	40.32 ↓	34.95 ↓	**40.61**↑	**21.49**↑	**30.18**↑	**29.31**↑
LLaVaNext-7B	**42.02**	**38.06**	40.05	19.78	28.10	26.54
LLaVaNext-13B + M-RAV	**43.02**↑	**42.48**↑	**41.48**↑	**36.61**↑	34.66 ↓	35.34 ↓
LLaVaNext-13B	40.07	34.24	41.29	25.21	**37.37**	**38.03**
LLaVaNext-34B + M-RAV	**50.07**↑	**49.15**↑	**45.94**↑	**38.99**↑	40.48 ↓	40.49 ↓
LLaVaNext-34B	45.42	41.59	43.64	30.35	**46.83**	**46.89**

Table 7
Comparison between M-RAV with SOTAs methods.

Dataset	Methods	F1-micro	F1-macro
Mocheg	Baseline (CLIP-large) (Yao et al., 2023)	50.78	–
	CLIP + GPT-4o Guided (Lee, et al., 2024)	54.22	–
	LVLM4FV (Tahmasebi, Müller-Budack, & Ewerth, 2024)	53.40	53.55
	M-RAV -*Qwen-2.5-32B* (our)	**57.37**	**56.77**
FACTIFY	Baseline (ResNET+SBERT) (Mishra, et al., 2022)	–	53.09
	Logically (Gao, Hoffmann, Oikonomou, Kiskovski, & Bandhakavi, 2021)	–	76.71
	ECENet (Zhang et al., 2023)	–	81.50
	M-RAV - *llama-3.1-70B* (our)	**72.33**	**74.87**
Fin-Fact	Baseline (Gemini Vision pro) (Rangapur et al., 2025)	58.00	59.00
	Baseline (LLaVA-13B) (Rangapur et al., 2025)	50.00	45.00
	FinLLaMA (Huang, Mengxi, et al., 2024)	34.62	–
	M-RAV - *llama-3.1-70B* (our)	**63.52**	**57.23**

the vision-language LLM LLaVA-NeXT (Li et al., 2024), as shown in Table 6. The details about the prompt for LLMs are shown in Appendix.

From Table 6, it can be seen that the M-RAV helps to increase the performance of LLaVA-34B and LLaVA-13B (vision-language LLMs) on the Mocheg dataset. However, on the Fin-Fact dataset, the M-RAV shows efficiency when it increases the accuracy of both text-only and vision-language LLMs (except LLaVANeXT-34B and Mixtral-8x-7b). The M-RAV especially increases the performance significantly of all LLMs on the FACTIFY dataset. In general, the M-RAV helps to improve the ability of LLMs to reason and understand the evidence context, which determines the truthfulness of the claim. In addition, as shown in Table 6, the performance of text-only LLMs such as LLaMA, Qwen, and Mixtral is better than vision-language LLMs like LLaVA. The *llama-3.1-70B* attains the best performance on both Fin-Fact and FACTIFY datasets when enhanced with M-RAV, while it performs well with Mocheg in the case of no M-RAV. On the other hand, the *Qwen-2.5-32B* obtains good results on the Mocheg when enhanced with M-RAV. Moreover, from Table 6, it can be seen that the M-RAV method shows a significant impact with 32 billion models or more since it leverages the ability of reasoning on large models for boosting the accuracy for the zero-shot fact verification task. Overall, the augmentation step, which integrates the explanation of the consistency between the image and the text evidence to the LLMs, helps improve the performance of the multimodal fact-checking task (answered for the **RQ1**). Besides, the quality of the evidence affects the performance of the fact-checking process by the M-RAV framework. When the evidence contains clear cross-modal information and is consistent with the claim, M-RAV performs well. In contrast, the performance of M-RAV is affected when the evidence is noisy or less similar to the claim.

Next, comparing the baseline shown in Table 7, our proposed method M-RAV shows the efficiency in boosting the LLMs for verifying the claim when the accuracy computed by F1 scores is much better than the results of baselines on three datasets. In the Mocheg dataset, the performance of *Qwen-2.5-32B* with M-RAV obtained 57.37% by F1-score, which is a better performance than CLIP+GPT-4o Guided with about 54.22% by F1-score (Lee, et al., 2024) and LVLM4FV with 53.40% by F1-score (Tahmasebi et al., 2024). Besides, the *llama-3.1-70B* with M-RAV also obtains potential results on Fin-Fact when compared with FinLLaMA, a series of LLMs trained on financial data (Huang, Mengxi, et al., 2024) (M-RAV with *llama-3.1-70B* obtained 74.87% by F1-score in comparison with FinLLaMA that attains 34.62%). On the FACTIFY dataset, the Logically (Gao et al., 2021) and ECENet (Zhang

Table 8
Ablation Study of M-RAV.

	Mocheg		FACTIFY		Fin-Fact	
	F1-micro	F1-macro	F1-micro	F1-macro	F1-micro	F1-macro
w/o relevance score	57.37	56.67	64.57	67.15	50.60	50.46
w/o alignment	47.32	47.30	69.41	72.46	31.49	29.64
w/o evidence	37.63	27.81	41.64	27.38	18.84	11.97
Full	57.37	56.77	66.86	70.13	49.33	48.60

et al., 2023) show high results since they were carefully trained and fine-tuned on the training set of FACTIFY. Moreover, the ECENet (Zhang et al., 2023) employed Deep Reinforcement Learning (DRL) to extract the most significant sentence that impacts the veracity of the claim and then adjust the weight through a reward model. From the empirical results, our M-RAV method generally yields competitive results with SOTAs, even without training or fine-tuning steps for the LLMs.

5.3. Ablation study

To investigate the impact of the relevance score, text-image alignment, and evidence in the M-RAV framework for fact-checking, we conducted an ablation study with three configurations: (1) excluding the relevance score (w/o relevance score), (2) excluding the text-image alignment while also omitting the relevance score (w/o alignment), and (3) using only the claim without supporting evidence (w/o evidence). For the "w/o alignment" configuration, it also removes the relevance score since the computation of the relevance score depends on the alignment stage between text and image evidence (as shown in Section 3). Table 8 presents the results of this ablation study, based on the *Qwen-2.5-32B-Instruct* model, evaluated across three benchmarks.

As shown in Table 8, alignment plays a crucial role in improving the verification model's performance on Mocheg and Fin-Fact, the two benchmarks that involve multiple pieces of evidence per claim. Without alignment, both macro and micro F1 scores drop significantly on these datasets. In contrast, alignment has minimal impact on FACTIFY, as this benchmark provides only a single piece of evidence — combining text and image — for each claim (as detailed in Section 4). Overall, the performance difference with and without alignment remains small. In practical scenarios with multiple evidence items per claim, aligning each text-image pair is essential to strengthening the verification model's ability to assess claim veracity. Furthermore, evidence is central to the verification process, as model performance declines sharply across all three benchmarks when evidence is absent.

Finally, the relevance score has less impact on the performance of verification models, as shown in Table 8. However, the relevance score plays a key role in filtering the most relevant alignment evidence for the claim, especially in a system-evidence scenario where the evidence is automatically retrieved. The next section presents an empirical study on system-evidence results, where the relevance score is used to identify the top-k most relevant alignments for the input claim.

5.4. System-evidence results

This section evaluates our M-RAV on different LLMs in the system evidence scenario. In this experiment, we only give the claim. The system retrieves the relevant evidence, including text and images, from the evidence database. Then, these pieces of evidence are augmented by M-RAV to extract the veracity of the claim.

In the retrieval step (Stage 1 of the framework as illustrated in Section 3), we employ the FlagEmbedding BGE-M3 (Chen, et al., 2024) for retrieving the text and CLIP (Radford, et al., 2021) for retrieving the image based on the claim as the query. We get the *top-k* most relevant evidence to the claim. To choose an efficient *top-k*, we test the performance of the retrieval model on *top-k* $\in \{5, 7, 10, 15, 20, 25, 30, 40, 50\}$, as shown in Table 9. We obtain that the ***top-5*** achieves the best performance according to the F1@k metric. Finally, we use the ***top-5*** for the retrieval model and attain the competitive results with the baseline in Mocheg (Yao et al., 2023), as shown in Table 10.

As outlined in the Evidence Alignment stage (Stage 2 of the framework that is described in Section 3), the alignment between text and image pairs explains their consistency. When selecting the top-N images and top-M texts, the total number of possible alignments is roughly $N \times M$. For example, choosing the top-5 images and top-5 texts generates 25 alignment pairs. This large number of pairs can create lengthy input for large language models (LLMs), potentially reducing their performance during claim verification. To address this issue, we conducted an empirical study to identify the most relevant text-image alignments using relevance scores from Stage 3 of the M-RAV framework (see Section 3). We sampled data from the Mocheg development set and used retrieval models to select the top-5 evidence items (both texts and images). Then, we calculated relevance scores for the alignments and selected the k most relevant pairs, where k was set to 5, 10, 15, 20, or 25. These pairs were used to verify claims with the *Qwen-2.5-32B-Instruct* model. As shown in Table 11, the top-5 alignments ($k = 5$) yielded the best performance. Therefore, for each claim, we use the top-5 most relevant text-image alignments based on their relevance scores.

Next, we augmented the retrieved evidence with the M-RAV method to get a list of verbalized evidence between text and image (as described in Fig. 2). Then, based on the relevance score with the claim, we choose *top-5* most relevant evidence to guide the LLMs in verifying the veracity. As shown in Table 12, the M-RAV boosted the performance of all text-only LLMs in the verification of the claim with system evidence. To the vision-language LLMs, the M-RAV enhanced the performance of LLaVaNext-34B. The Qwen-2.5-32B obtains the best performance with 50.02% by F1-micro score, which is approximately 6% better than the baseline, and

Table 9
Performance of the retrieval module on the development set of Mocheg by different top-k.

top-k	Modality	prec@K	rec@K	F1@K	MAP
k=5	Image	**7.50**	27.13	**11.35**	22.47
	Text	**23.70**	51.46	**30.19**	**57.86**
k=7	Image	5.76	29.22	9.35	22.65
	Text	18.88	56.21	26.48	57.17
k=10	Image	4.25	31.13	7.31	22.80
	Text	14.23	59.80	21.75	56.58
k=15	Image	3.09	34.61	5.60	22.71
	Text	10.04	63.15	16.59	55.68
k=20	Image	2.44	36.77	4.52	22.81
	Text	7.79	65.07	13.43	55.04
k=25	Image	2.02	38.08	3.80	22.84
	Text	6.39	66.46	11.31	54.42
k=30	Image	1.74	39.54	3.32	22.81
	Text	5.42	67.58	9.78	54.14
k=40	Image	1.35	41.12	2.61	22.78
	Text	4.19	69.60	7.74	53.46
k=50	Image	1.10	41.75	2.14	22.78
	Text	3.42	70.73	6.42	52.97

Table 10
Result of retrieval module on the test set of Mocheg.

Method	Modality	prec@K	rec@K	F1@K	MAP
BGE+CLIP (k=5)	Image	4.26	14.89	6.31	**12.32**
	Text	**22.45**	**47.68**	**29.06**	**58.71**
Baseline (k=5)	Image	4.71	17.01	7.37	11.93
	Text	*14.92*	*19.72*	*16.98*	14.34
Baseline (k=10)	Image	3.02	21.44	5.29	12.58
	Text	*9.79*	*23.99*	*13.90*	15.34

Table 11
The impact of filtering the top-k most relevant evidence alignment pairs for verifying the claim.

top-K alignment	F1-micro	F1-macro
5	**49.00**	**49.42**
10	46.83	45.16
15	45.83	41.87
20	45.83	40.20
25	44.50	38.95

slightly reaches the performance of the baseline trained on gold evidence of Mocheg (Yao et al., 2023). Therefore, the results from the empirical study indicate that the M-RAV is efficient for applying to the fact-checking system, where the evidence is automatically retrieved (answered for the **RQ2**).

6. Error analysis and discussion

Table 13 describes the detailed performance of the classification of LLMs on the three datasets. For text-only LLMs, we choose the results of the model that has the best performance on each dataset by M-RAV according to Tables 6 and 12, in which the *Qwen2.5-32B* for Mocheg, the *llama-3.1-70B* for FACTIFY, and Fin-Fact. Similarly, we choose the *LLaVaNext-34B* as the vision-language LLM for Mocheg, FACTIFY, and Fin-Fact.

In general, it can be seen that text-only LLMs have better performance than vision-language LLMs. Although the efficiency of M-RAV is faint on text-only LLMs, it is significant on vision-language LLMs since it helps improve performance. Especially, on the Mocheg with system evidence, the M-RAV helps improve almost both text-only and vision-language models, indicating the

Table 12
Empirical results on system evidence.

Method	F1-micro	F1-macro
llama-3.2-3B + M-RAV	**32.63**↑	28.34 ↓
llama-3.2-3B	31.48	**29.96**
llama-3.1-8B + M-RAV	**49.02**↑	**41.22**↑
llama-3.1-8B	45.32	42.66
Qwen-2.5-32B + M-RAV	**50.02**↑	**50.14**↑
Qwen-2.5-32B	42.62	38.06
llama-3.1-70B + M-RAV	**49.17**↑	**46.33**↑
llama-3.1-70B	45.07	43.55
Mixtral-8x-7b + M-RAV	**49.52**↑	**47.55**↑
Mixtral-8x-7b	47.42	45.66
LLaVaNext-7B + M-RAV	33.28 ↓	16.74 ↓
LLaVaNext-7B	**33.38**	**16.79**
LLaVaNext-13B + M-RAV	33.23 ↓	16.73 ↓
LLaVaNext-13B	**33.43**	**16.90**
LLaVaNext-34B + M-RAV	**38.73**↑	**30.91**↑
LLaVaNext-34B	33.38	21.17
Baseline (Yao et al., 2023)	44.06	–

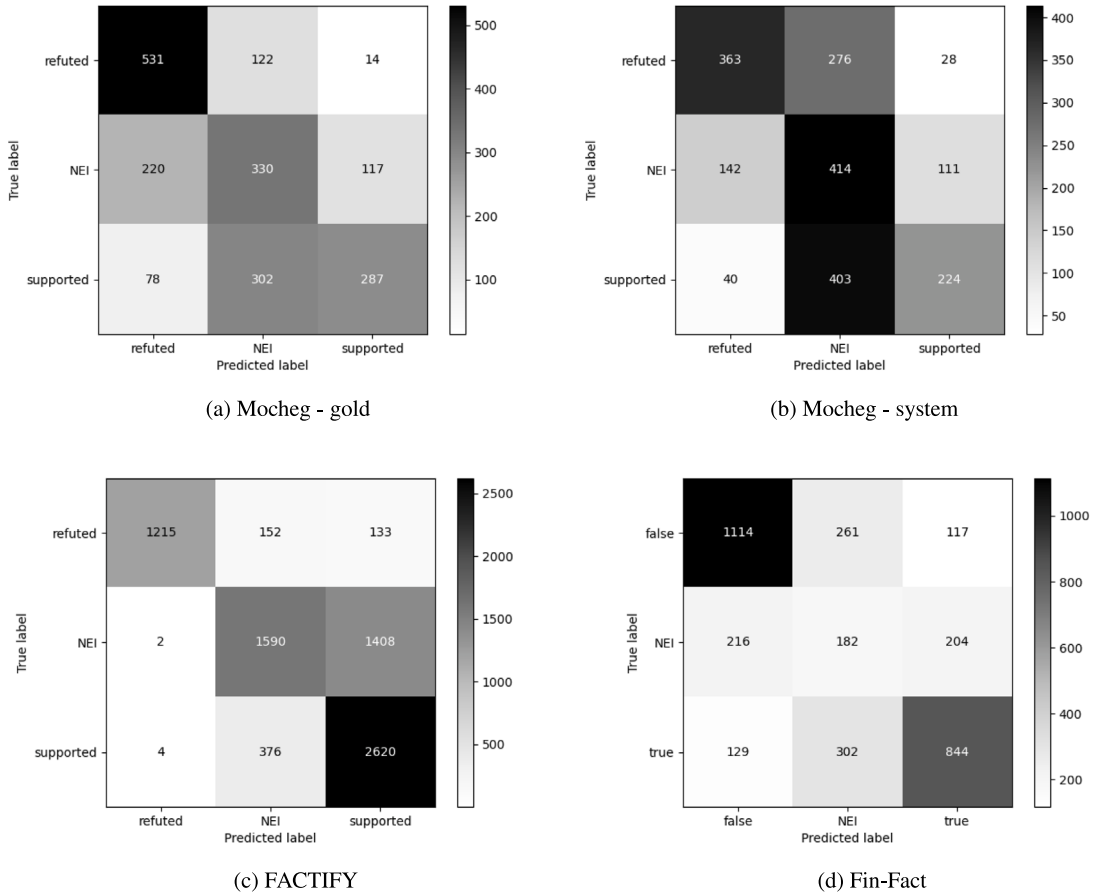
Table 13
Performance results of text-only and vision-language LLMs on the three datasets. F1 represents the F1-macro score, P is Precision, and R is Recall.

	Macro F1	*Macro P*	*Macro R*	*Support F1*	*Support P*	*Support R*	*NEI F1*	*NEI P*	*NEI R*	*Refute F1*	*Refute P*	*Refute R*
	Mocheg - gold evidence											
Text-only LLMs + M-RAV	**56.78**	58.83	**57.37**	**68.66**	52.90	**43.03**	70.99	64.05	**79.61**	46.45	43.77	49.48
Text-only LLMs	49.83	**59.48**	53.47	26.94	**70.70**	16.64	**71.99**	**66.54**	78.41	**50.55**	41.21	**65.37**
Vision-language LLMs + M-RAV	**49.16**	57.16	**50.07**	**62.11**	53.40	**74.21**	**45.12**	81.57	**31.18**	40.24	36.51	44.83
Vision-language LLMs	41.59	**63.00**	45.43	28.50	**68.57**	17.99	43.93	**83.26**	29.84	**52.35**	**37.18**	**88.46**
	Mocheg - system evidence											
Text-only LLMs + M-RAV	**50.15**	**55.40**	**50.02**	**43.50**	**61.71**	33.58	**59.90**	**66.61**	**54.42**	**47.05**	**37.88**	62.07
Text-only LLMs	38.06	56.24	42.63	23.92	67.36	14.54	39.50	65.17	28.34	50.76	36.18	**85.01**
Vision-language LLMs + M-RAV	**30.92**	**58.63**	**38.73**	**54.12**	**43.20**	**72.41**	**1.49**	**100**	**0.75**	**37.15**	**32.69**	43.03
Vision-language LLMs	21.18	35.00	33.38	14.93	43.80	0.90	0.59	28.57	0.30	48.02	32.63	**90.85**
	FACTIFY											
Text-only LLMs + M-RAV	**74.87**	**79.18**	**73.78**	**73.17**	62.97	**87.33**	89.31	**99.51**	81.00	62.13	**75.07**	53.00
Text-only LLMs	74.44	77.13	72.93	70.06	**64.01**	77.37	**89.52**	99.35	**81.47**	**63.75**	68.04	**59.97**
Vision-language LLMs + M-RAV	**38.99**	61.78	**40.44**	**59.25**	48.71	**75.60**	**22.89**	**99.49**	**12.93**	34.84	37.15	32.80
Vision-language LLMs	30.36	**68.45**	38.28	12.75	**65.23**	7.07	20.55	98.85	11.47	**57.77**	**41.27**	**96.30**
	Fin-Fact											
Text-only LLMs + M-RAV	57.23	57.74	57.03	69.18	72.45	**66.20**	**75.50**	76.35	**74.66**	27.02	24.43	30.23
Text-only LLMs	**59.62**	**64.04**	**61.82**	**69.29**	**73.46**	65.57	69.04	**88.91**	56.43	**40.51**	**29.75**	**63.46**
Vision-language LLMs + M-RAV	40.49	49.22	43.49	**49.85**	54.91	**45.65**	43.09	73.39	30.50	28.55	19.36	54.32
Vision-language LLMs	**46.90**	**56.26**	**49.73**	49.05	**68.01**	38.35	**59.24**	**78.99**	**47.39**	**32.41**	**21.77**	**63.46**

contribution of M-RAV in a practical fact-checking system when the evidence is retrieved automatically instead of manually created as gold evidence.

Next, from Table 13, the Mocheg (both gold and system evidence) is mostly misclassified on the Refute and NEI labels. As shown by the confusion matrices in Figs. 9,10,11,12, most of the time the Refutes is predicted as NEI, while the NEI tends to be predicted as Support. For the FACTIFY dataset, the Refuted model is mostly misclassified to the NEI label and Support label on text-only LLMs, while it is focused mostly on the NEI label when testing with vision-language LLMs. Finally, on the Fin-Fact dataset, the Refute label tends to be predicted as NEI. Overall, it can be seen that the Support and Refute are clearly discriminated against, while the remaining challenge falls into the NEI labels. To investigate more about the error, we categorize the misclassification sample into 4 types: **Error type 1:** The "support" is being predicted as "NEI", **Error type 2:** The "refute" is being predicted as "NEI", **Error type 3:** The "NEI" is being predicted as "support", and **Error type 4:** The "NEI" is being predicted as "refute".

To better visualize the error prediction, we extracted several error prediction samples. For **Error type 1**, it can be seen that the model cannot capture the latent information from the image and text. For example, in Fig. 13, the LLMs understand the *"tall*



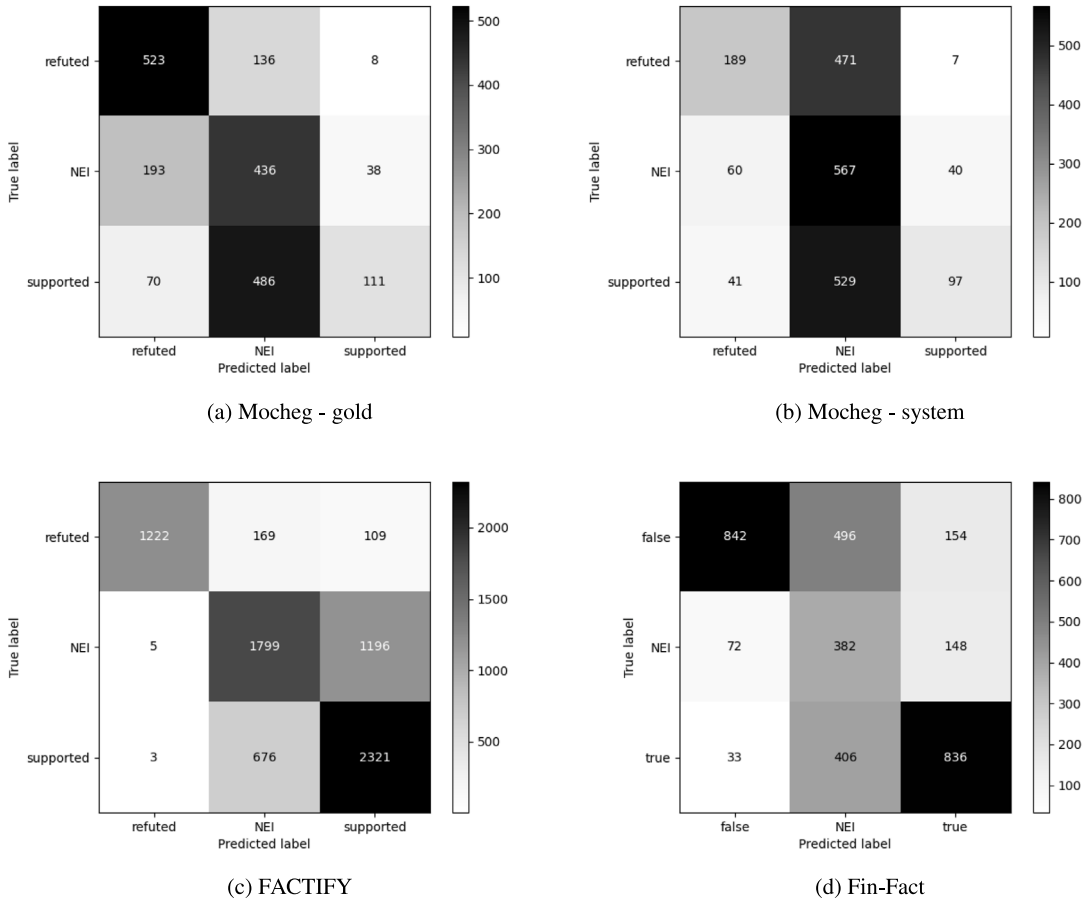
(a) Mocheg - gold (b) Mocheg - system

(c) FACTIFY (d) Fin-Fact

Fig. 9. Performance of text-only LLMs with M-RAV.

building" and the *"Burj Khalifa"* in the explanation between the image and the text. Besides, it can exploit the information as *"a person working on a tall building with a cityscape in the background"* from the explanation between the consistency of the image and text. However, the model does not understand *"Tom Cruise"* as a person in the picture thus, it cannot determine whether *"Tom Cruise"* sits on the top of *"a tall building"* named *"Burj Khalifa"*. Therefore, the model predicted "NEI" while the correct label is "support". Next, in Fig. 14, the LLM can understand what "Pokemon" is as well as exploit correct contextual information about the "Pokemon" such as *"Pikachu balloon"* or *"Pikachu costume"*. However, the model does not understand *"witchcraft"* (mentioned in the claim) and *"satanic ritual"* (mentioned in the first text evidence) since it predicts "NEI". Besides, it can be seen from Fig. 15 (an example for **Error type 2**) that in the explanation, the information about *"a group of sheep passing through a gate"* and *"a fence in the background"* has been correctly extracted. With the first piece of evidence, it is enough to determine the truthfulness of the claim as "refuted" because the explanation and the text image said that *"sheep passing through a gate at Brenda Station"* and *"the image accurately represents the scene described in the text"* is matching with the information in the text evidence as *"fence posts are clearly visible in a higher resolution version"*. However, in the second piece of evidence, there is a little confusion since the information in the image shows that *"fence posts visible in the background"*, which contradicts the information in the text evidence as *"encircling fence may be hard to see"*. Thus, the model predicts the claim as "NEI" instead of "Refuted".

In addition, from Fig. 16, the text evidence at first seems to support the claim since it says that *"The Tripura government will promote all students from Class 1 to 4 and Class 6 and 7 without examinations this year. Ẅe have decided to promote students from classes 1 to 4, 6, and 7 to the next classes except Classes 5 and 8"*.. Also, the image is consistent with the text as the explanation shows the information about the *"discussion of promoting students without examinations and extending summer vacations"*. However, in the text evidence, the remaining information seems uncertain about whether the government decided for *"No examination for Class 1 to 4 and Class 6 to 7"* because the text evidence mentioned that *"But the students need to sit for their examinations after opening of schools if*



(a) Mocheg - gold

(b) Mocheg - system

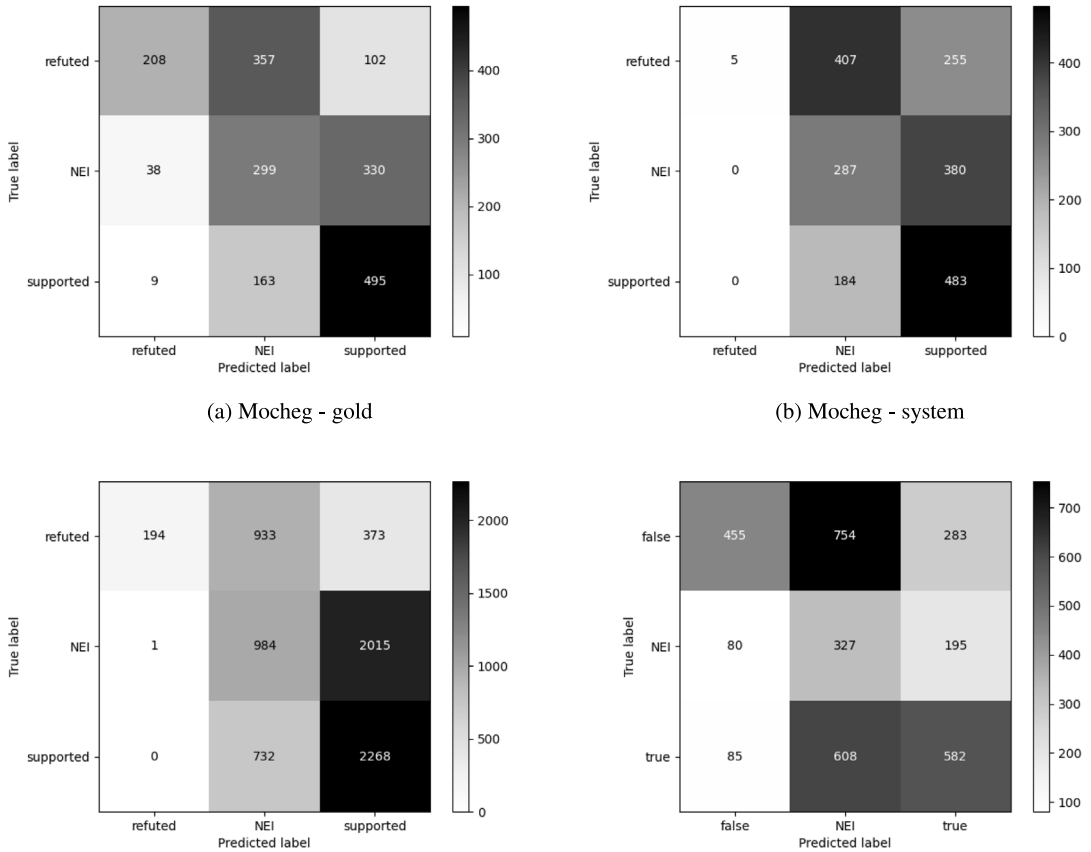
(c) FACTIFY

(d) Fin-Fact

Fig. 10. Performance of text-only LLMs without M-RAV.

the situation becomes normal".. Moreover, from the evidence, it is said that *"The exams would be held to evaluate their educational loss in the Covid-19 pandemic"* and *"Regarding Class 5 and 8, Nath said that they would send the matter for approval from the state cabinet and for Class 9 and 11 examinations"*, which means it is not sure that *Class 1 to 4 and Class 6 to 7* do not have to take the exam, and *Class 5 and Class 8* must have to take the exam. Therefore, the actual truthfulness of this claim is "NEI". Nevertheless, the model was predicted as "support", which falls into **Error type 3**. Finally, for the sample of **Error type 4** shown in Fig. 17, it can be seen that, as with the evidence extracted from the image via explanation and from the text evidence, the claim must be refuted. However, no information showed that the *"Facebook user Rosemary Thomas"* is a customer who found the Rat Head in the food from Popeyes Chicken since this is *"the iteration of a very old urban legend known as the Kentucky Fried Rat"* . Consequently, the actual label of the claim is "NEI".

Overall, it can be seen that the evidence in some cases is not enough to determine the veracity of the claim, even if it is correct and clearly represented. To exactly verify the claim, the LLM needs external knowledge about specific cultures, politics, and human behavior to capture the latent contextual information along with the evidence in verifying the claim. To improve how LLMs adapt to new information, several techniques can be used. CAMeL (Yu, Wen, & Zheng, 2025) and MemoryLLM (Wang et al., 2024) help LLMs integrate new knowledge into their internal parameters. Alternatively, approaches like LLatrieval (Li, et al., 2024) enable LLMs to iteratively validate external knowledge, enhancing their ability to verify information and augment their performance.



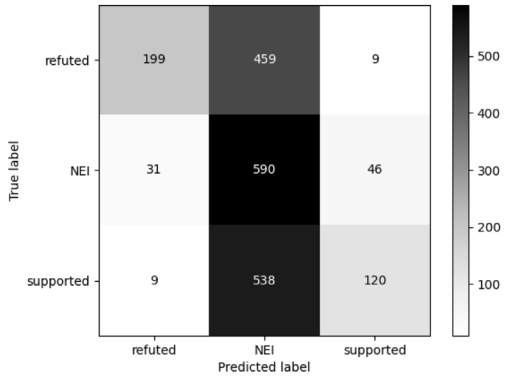
(a) Mocheg - gold

(b) Mocheg - system

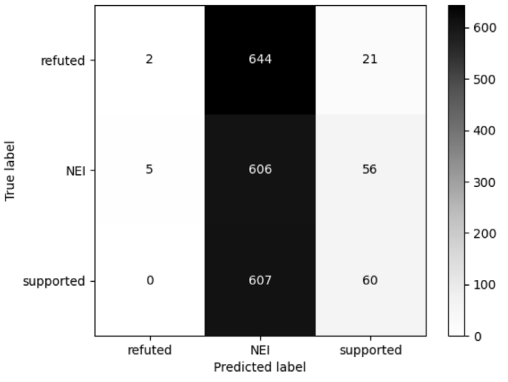
(c) FACTIFY

(d) Fin-Fact

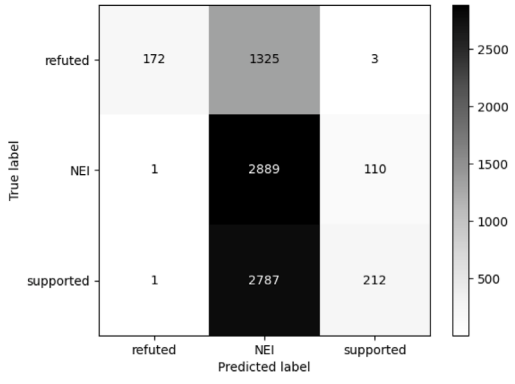
Fig. 11. Performance of vision-language LLMs with M-RAV.



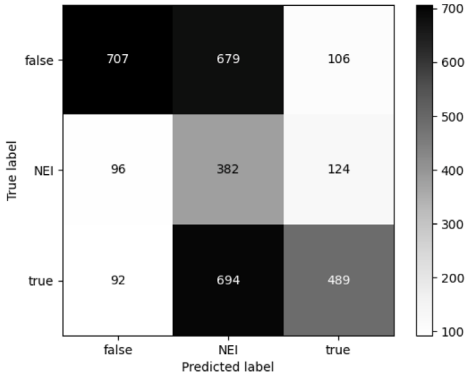
(a) Mocheg - gold



(b) Mocheg - system



(c) FACTIFY



(d) Fin-Fact

Fig. 12. Performance of vision-language LLMs without M-RAV.

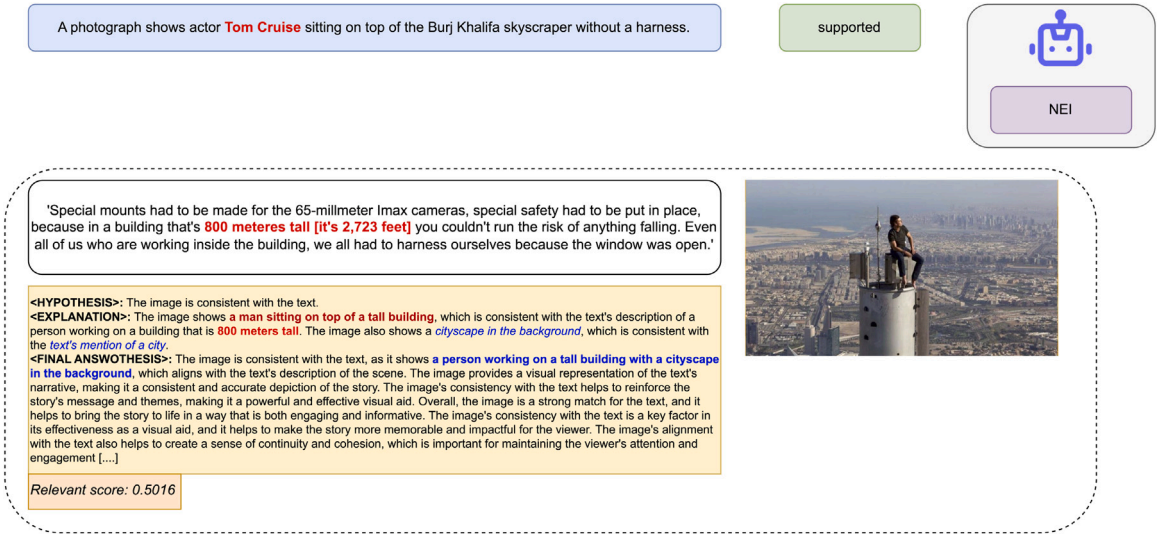


Fig. 13. Error prediction sample 1. The sample is extracted from the Mocheg dataset.

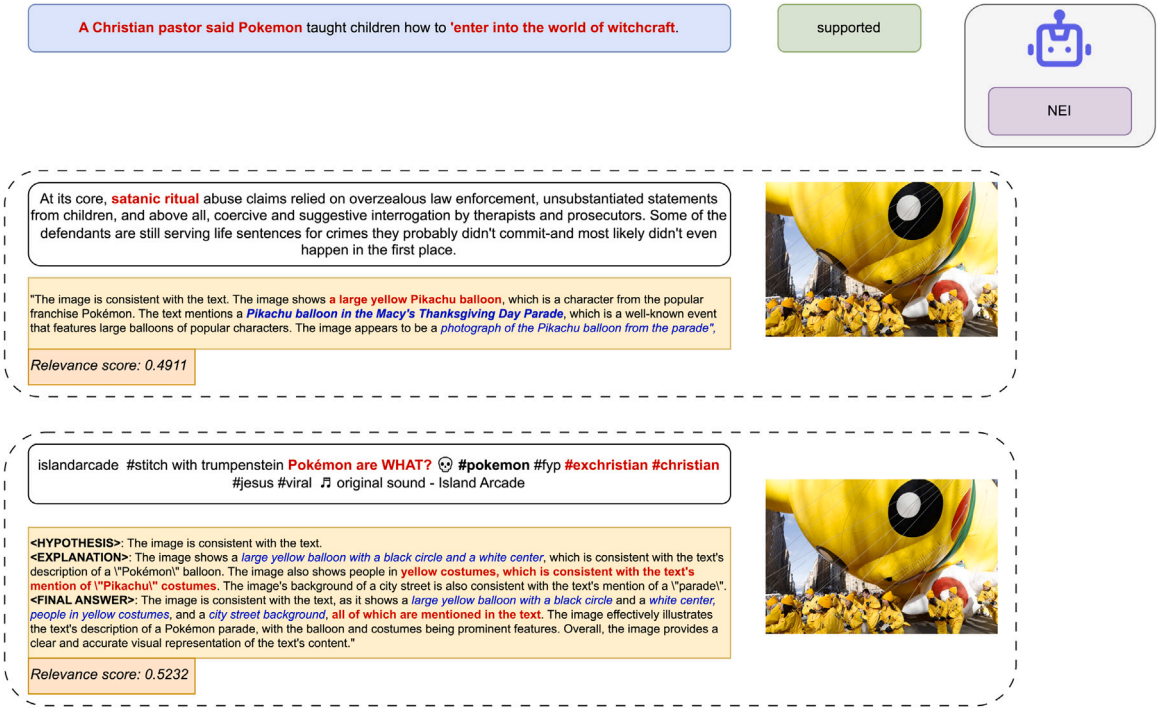


Fig. 14. Error prediction sample 2. The sample is extracted from the Mocheg dataset.



Fig. 15. Error prediction sample 3. The sample is extracted from the Fin-Fact dataset.

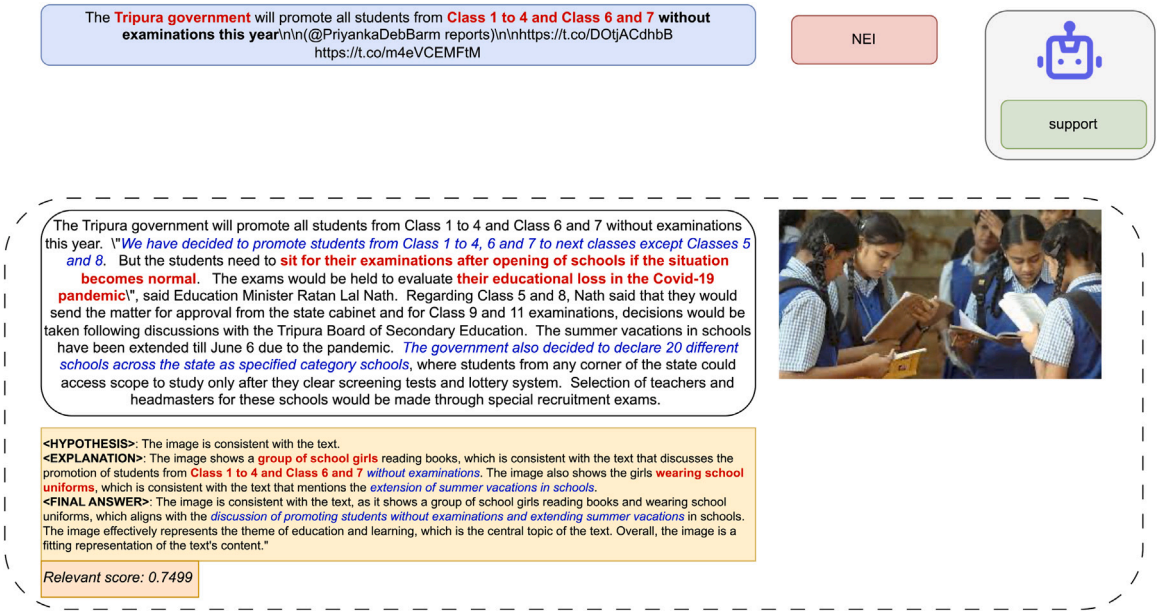


Fig. 16. Error prediction sample 4. The sample is extracted from the FACTIFY dataset.

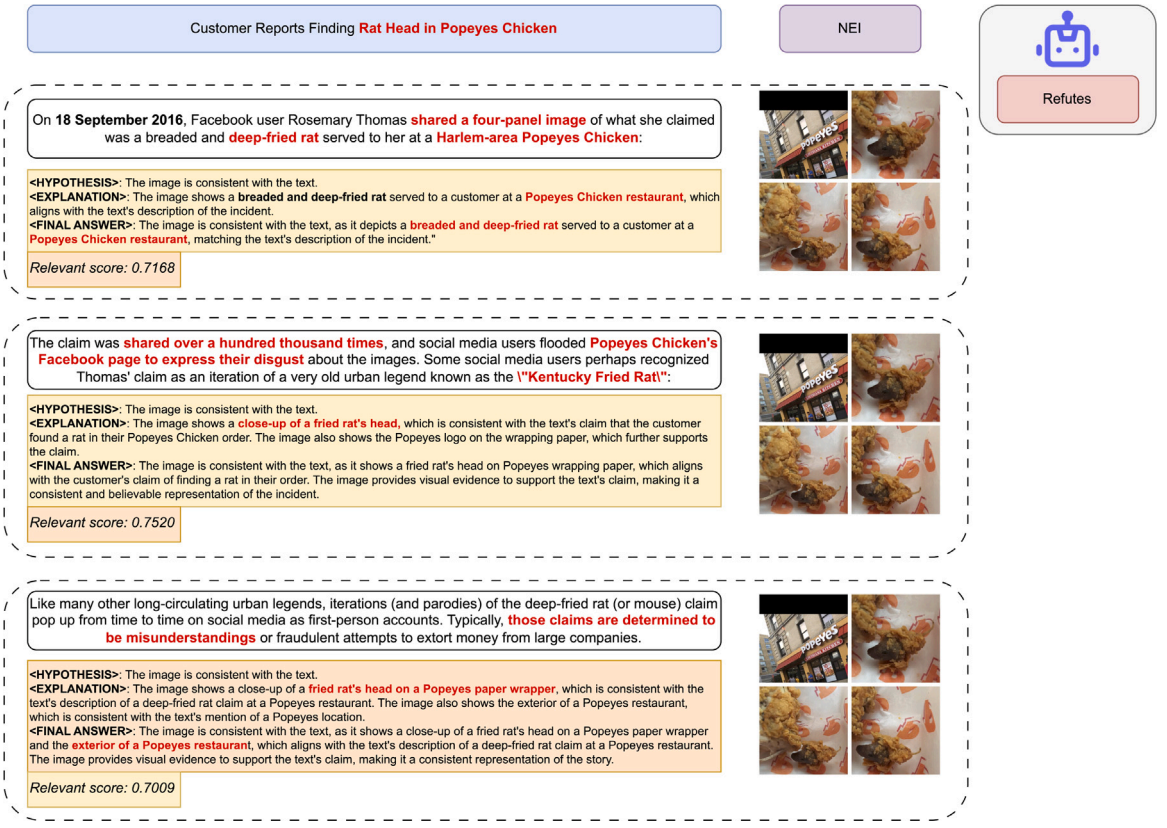


Fig. 17. Error prediction sample 5. The sample is extracted from the Fin-Fact dataset.

7. Conclusion

We propose M-RAV, a **M**ultimodal **R**etrieve-**A**ugment-**V**erify framework designed to enhance the ability of large language models (LLMs) to understand evidence and verify the truthfulness of claims. Our framework significantly improves the performance of open-source text-only and vision-language LLMs in zero-shot fact-checking tasks. Empirical results demonstrate that M-RAV enables open-source LLMs to surpass baseline models and achieve performance comparable to state-of-the-art (SOTA) models, such as commercial LLMs and fine-tuned models on specific datasets. Moreover, M-RAV proves effective in practical fact-checking systems by automatically retrieving evidence, eliminating the need for manually provided ground-truth evidence. These findings highlight M-RAV's potential for robust multimodal fact-checking in real-world scenarios.

Error analysis reveals that LLMs struggle to verify claims labeled as "not enough information" (NEI), even when sufficient evidence is provided. This challenge arises because LLMs often fail to interpret latent social factors, such as cultural or political contexts, which are critical for accurate verification. To address this limitation, our future work will explore techniques like MemoryLLM (Wang et al., 2024) and MemoryBank (Zhong, Guo, Gao, Ye, & Wang, 2024) to dynamically update LLMs' knowledge bases with external environmental information or iteratively validate new external knowledge like LLatrieval (Li, et al., 2024). These advancements aim to enhance LLM performance in fact-checking and other complex tasks requiring contextual understanding.

Additionally, we plan to expand the evaluation of M-RAV across a broader range of robust LLMs, as well as integrate human-in-the-loop for Fact-checking system development. Our goal is to develop a comprehensive end-to-end fact-checking system with seamless M-RAV integration, further improving its applicability and effectiveness in practical settings.

CRediT authorship contribution statement

Son T. Luu: Writing – original draft, Software, Methodology, Formal analysis. **Trung Vo:** Writing – review & editing, Visualization, Validation. **Le-Minh Nguyen:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

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Appendix. Prompt for LLMs with M-RAV

A.1. Prompt for text-only LLMs with sufficient text and image

This sample is from FACTIFY with *llama3.1-70B*.

```
Is it true that: The Tripura government will promote all students from Class 1 to 4 and Class 6 and 7 without
    examinations this year\n\n(@PriyankaDebBarm reports)\n\nhttps://t.co/DOtjACdhbB https://t.co/
    m4eVCEMFtM?
The evidence:
      Evidence 1:
      Text: The Tripura government will promote all students from Class 1 to 4 and Class 6 and 7 without
    examinations this year. \"We have decided to promote students from Class 1 to 4, 6 and 7 to next classes
    except Classes 5 and 8. But the students need to sit for their examinations after opening of schools if
    the situation becomes normal. The exams would be held to evaluate their educational loss in the Covid-19
    pandemic\", said Education Minister Ratan Lal Nath. Regarding Class 5 and 8, Nath said that they would
    send the matter for approval from the state cabinet and for Class 9 and 11 examinations, decisions would
    be taken following discussions with the Tripura Board of Secondary Education. The summer vacations in
    schools have been extended till June 6 due to the pandemic. The government also decided to declare 20
    different schools across the state as specified category schools, where students from any corner of the
    state could access scope to study only after they clear screening tests and lottery system. Selection of
    teachers and headmasters for these schools would be made through special recruitment exams.

      Image: The image shows a group of school girls reading books, which is consistent with the text that
    discusses the promotion of students from Class 1 to 4 and Class 6 and 7 without examinations. The image
    also shows the girls wearing school uniforms, which is consistent with the text that mentions the
    extension of summer vacations in schools. The image is consistent with the text, as it shows a group of
    school girls reading books and wearing school uniforms, which aligns with the discussion of promoting
    students without examinations and extending summer vacations in schools. The image effectively
    represents the theme of education and learning, which is the central topic of the text. Overall, the
    image is a fitting representation of the text's content.
      Consistency: The image is consistent with the text.
      Relevance score: 0.7499760389328003
```

```
To verify the truthfulness of the claim, please follow these steps:
    STEP 1: Consult the relevance between the claim and each given evidence based on the relevance score.

       STEP 2: Think and conclude the truthfulness of the claim based on the relevance and logic of the
     evidence. If the evidence does not help conclude whether the claim is supported or refuted, it may be not
      enough information.

The truthfulness must be only one of three values: supported, refuted, or not enough information. Please
     think step-by-step carefully and response only the truthfulness of the claim.

<RESPONSE>:
```

A.2. Prompt for text-only LLMs without image evidence

This sample is from Mocheg with *mixtral-8x-7b*.

```
Is it true that: Image depicts a huge, inexplicably half-eaten shark found on a Florida beach.?
The evidence:
       Evidence 1:
       Text: People taking surf lessons on Butler Beach were in a for big surprise when they entered the
     water Saturday. At around 12:30 p.m., Dakota Dodson, a surf instructor for the St. Augustine Surf School
     , noticed a fin sticking out of the water and went to investigate. 'At first, I only saw the fin, so I just
      assumed it a live shark. But then I realized it was only half of one,' he said. The half-eaten carcass,
     measuring about 2.5 feet, was found floating in knee-deep water, Dodson said. The circumstances of how
     the shark died remain unclear, but experts believe it was likely eaten by another shark. 'It's most
     likely it was bitten in half by another shark - sharks have to eat, too,' said Tara Dodson, environmental
      supervisor for St. Johns County.
       Relevance score: 0.5962506532669067

To verify the truthfulness of the claim, please follow these steps:
       STEP 1: Consult the relevance between the claim and each given evidence based on the relevance score.
       STEP 2: Think and conclude the truthfulness of the claim based on the relevance and logic of the
     evidence. If the evidence does not help conclude whether the claim is supported or refuted, it may be not
      enough information.
The truthfulness must be only one of three values: supported, refuted, or not enough information. Please
     think step-by-step carefully and response only the truthfulness of the claim.
<RESPONSE>:
```

A.3. Prompt for vision-language LLMs

This sample is from FINFACT with *llava1.6-34B*. In the Tokenizer, the LLaVA accepts multiple images as a list of sequences since the *<image>* tokens in the prompt correspond to each image in the evidence.

```
Is it true that: 'So Far No One Has Found Another Number' Walmart Gift Card Facebook Scam? Here are the
     evidence for checking:
       Evidence 1 :
              Text: In late July 2023, readers pointed us to several posts that had been made inside of
     Facebook groups. The users who made these posts, whose profiles all indicated they may have been from
     Bangladesh, claimed to be offering Walmart gift cards to anyone who could find a special number in an
     image. For example, one post read, \"So Far No One Has Found Another Number Apart From 86, No Winners Yet
     (Walmart Gift Card). We Still Have 24 More Wins. \"\n
              Image: <image>
              Description: The image is consistent with the text, as it shows a hand holding a Walmart gift
     card and a grid of numbers, which aligns with the text's mention of a Walmart gift card and finding a
     number. The image and text also have a similar tone and style, with a playful and promotional feel.
     Overall, the image effectively supports the text and helps to convey the message in a visually engaging
     way.
              Consistency: The image is consistent with the text.
              Relevance score: 0.6894003748893738
       Evidence 2 :
              Text: Survey scam websites usually promise cash prizes, pricey electronics, and other
     interesting purported \"rewards, \" all supposedly if the user takes a few minutes to answer some
     questions. However, as we have reported for the last two decades or so, survey scam websites have
     historically proven to be a waste of time. They often ask users to provide personal and financial
     information on various websites, as well as to sign up for trials of unfamiliar streaming services. All
     of this appeared to be an attempt at receiving affiliate-marketing commission based on the amount of
     information given away when providing personal and financial data to these websites.
```

```
        Image: <image>
        Description: The image is consistent with the text, as it shows a Facebook post with a photo of
    a Walmart gift card and a caption that reads, \"So Far No One Has Found Another Number Apart From 86, No
    Winners Yet (Walmart Gift Card). We Still Have 24 More Wins. \" The image also includes a yellow circle
    with a white \"S\" inside it, which is likely a logo for a scam alert. This suggests that the image is a
    screenshot of a Facebook post that is warning people about a scam. Therefore, the image is consistent
    with the text.
        Consistency: The image is consistent with the text.
        Relevance score: 0.6600311994552612
  Evidence 3 :
        Text: Sometimes, these kinds of scammers might instead provide a link sending users to hidden
    subscription scams that supposedly offer \"free\" prizes. However, such scams hide monthly fees in the
    fine print, much like a Cash App scam we once reported about.
        Image: <image>
        Description: The image is consistent with the text, as it shows a Facebook post from Isabelle
    Freya, a scammer, and the text is a scam. The image and text are consistent because they both have the
    same scammer and the same scam. This suggests that the image and text are likely to be from the same
    source and are intended to deceive users. The consistency between the image and text highlights the
    importance of being cautious when encountering suspicious content online. It is essential to verify the
    authenticity of information and be aware of potential scams to avoid falling victim to fraudulent
    activities. By recognizing the consistency between the image and text, users can take steps to protect
    themselves from scams and maintain a safe online experience.
        Consistency: The image is consistent with the text.
        Relevance score: 0.5112748146057129
  Evidence 4 :
        Text: These sorts of scammers might also sometimes direct users to phishing websites that
    claim a gift card or other prize could be ordered for \"free, \" only with a small shipping and handling
    charge. Of course, there would be no real gift card or other prize. This simply would be an attempt to
    obtain a victim's financial information for criminal activities, such as a credit card number or PayPal
    login. This kind of a scam was similar to another one we previously reported about concerning the U. S.
    Postal Service.
        Image: <image>
        Description: The image is consistent with the text, as it shows a Facebook post from Isabelle
    Freya that matches the text in the image. The image also shows a hand holding a Walmart gift card and a
    grid of numbers with the number 86 repeated multiple times, which is consistent with the text in the post
    . Overall, the image appears to be a screenshot of the Facebook post, and it is consistent with the text.
        Consistency: The image is consistent with the text.
        Relevance score: 0.6907112002372742
  Evidence 5 :
        Text: If readers are looking for legitimate promotions for Walmart gift cards, we recommend
    our previous reporting that found the company truly does give away $80,000 in prizes every three months.
    Such promotions are offered by the company in official email correspondence and on receipts handed out
    in its brick-and-mortar stores.
        Image: <image>
        Description: The image is consistent with the text, as it shows a Facebook post from Isabelle
    Freya that mentions a Walmart gift card and the number 86, which is also depicted in the image. The image
    also includes a yellow \"SCAM\" logo, which suggests that the post may be a scam. Overall, the image and
    text are consistent in their content and message.
        Consistency: The image is consistent with the text.
        Relevance score: 0.6813938617706299
To verify the truthfulness of the claim, please following these steps:
    STEP 1 : Consult the relevance between the claim and each given evidence based on the relevance score.
    STEP 2 : Think and conclude the truthfulness of the claim based on the relevance and logic of the
    evidence.
The truthfulness must be only one of three values: supported, refuted, or not enough information. Please
    think step-by-step carefully and response only the truthfulness of the claim.
<RESPONSE>:
```

A.4. Prompt for vision-language LLMs to generate an explanation for the consistency between text and image evidence

This sample is from the Mocheg dataset for generating the consistency explanation between the text and image evidence. The *<image>* token in the prompt corresponds to the image evidence. We used the *Llama-3.2-90B-Vision-Instruct* for generation.

```
<|image|>
Agriculture Secretary Tom Vilsack today released the following statement regarding the language in the
    omnibus bill repealing the country of origin labeling requirements for beef and pork products. 'The
    omnibus bill repealed the country of origin labeling (COOL) requirements for muscle cuts of beef and
    pork, and ground beef and pork. Effective immediately, USDA is not enforcing the COOL requirements for
    muscle cut and ground beef and pork outlined in the January 2009 and May 2013 final rules.' USDA will be
    amending the COOL regulations as expeditiously as possible to reflect the repeal of the beef and pork
    provisions. In addition, all imported and domestic meat will continue to be subject to rigorous
    inspections by USDA to ensure food safety.

Please generate a short paragraph describing the consistency of the image based on the given text, following
    this template:
<HYPOTHESIS>: Please determine whether the image is consistent with the text or not.
<EXPLANATION>: Explain the alignment between the image hypothesis and the text.
<FINAL ANSWER>: Give one paragraph describing the consistency of the image and text based on the explanation
    .

<RESPONSE>
```

Data availability

The Mocheg dataset is available at this link: https://github.com/VT-NLP/Mocheg. The FACTIFY dataset is available at https://github.com/Shreyashm16/Factify. The Fin-Fact is available at https://github.com/IIT-DM/Fin-Fact

The source code for M-RAV is available at: https://github.com/sonlam1102/m-rav .

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